MILITARY SPECIFICATION

FILTER ELEMENT, FLUID, PRESSURE MXU-233/E

1. SCOPE

1.1 This specification covers one type of pressure fluid filter element, designated MXU-233/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	Box, Setup
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, Performance Requirements and Test Procedures for
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

D 256	*Standard Methods of Test for Impact Resistance of Plastics and Electrical Insulating Materials*
D 638	Method of Test for Tensile Properties of Plastics
D 790	Standard Method of Test for Flexural Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-F-8901 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Metallic parts. All metallic parts of the filter element shall be fabricated from an aluminum alloy suitable for the application.

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3.4.2 Fungusproof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.4.3 Petroleum products. The materials used in the construction of the filter element shall neither affect nor be affected by petroleum products.

3.4.4 Plastic parts. If used in the construction of the filter element, molded plastic parts shall have the following physical properties:

a. Tensile strength minimum - 4,500 psi

b. Flexural strength minimum - 7,500 psi

c. IZOD impact minimum 1.5 pounds per inch of notch.

3.5 Design and construction. The filter element shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

3.5.1 Functional design. The filter element shall be designed for installation in the Warner-Lewis Division of Fram Corporation filter-separators specified in 6.1.1. The filter element shall be as shown on figure 1.

3.5.1.1 Compatible fuels. The filter element shall be so designed that it can be used with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.5.1.2 Gasket. A gasket shall be furnished with each filter element. The gasket shall have an inside diameter of 2-1/4 inches, an outside diameter of 3-1/2 inches, and shall be 1/8 inch thick. The material for the gasket shall be Buna N or other material compatible with turbine fuels conforming to MIL-T-5624.

3.6 Performance. When 10 elements are installed in a Warner-Lewis Division of Fram Corporation model FCS 105 filter-separator, the elements shall be capable of meeting the performance requirements of MIL-F-8901 at a flow rate of 100 gpm, except the maximum water removal capability shall be 5 percent by volume.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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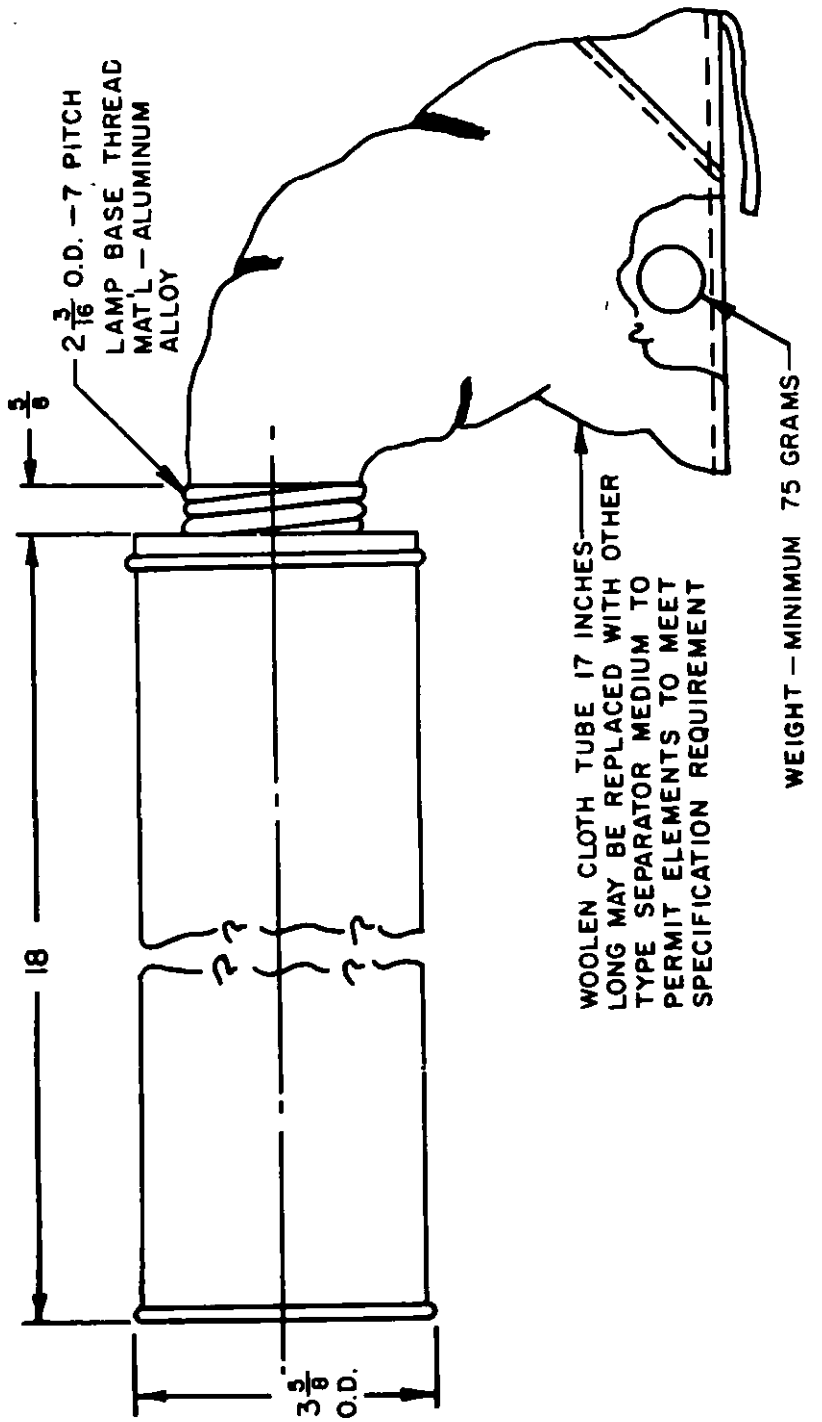


FIGURE 1. Filter Element

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3.8 Identification of product. The following information shall be marked on each filter element in accordance with MIL-STD-130. If the item is marked on any sealing surface, the markings shall not affect the sealing of the element.

Manufacturer's part No._____

Manufacturer's name or trademark_____

Date of manufacture_____.

3.8.1 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

3.8.2 Nameplate. A nameplate conforming to type I of MIL-P-19834 shall be supplied one per each 10 elements shipped. Attaching instructions shall be affixed to each nameplate. The nameplate shall conform to figure 2.

3.9 Workmanship. The filter element, including all parts, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

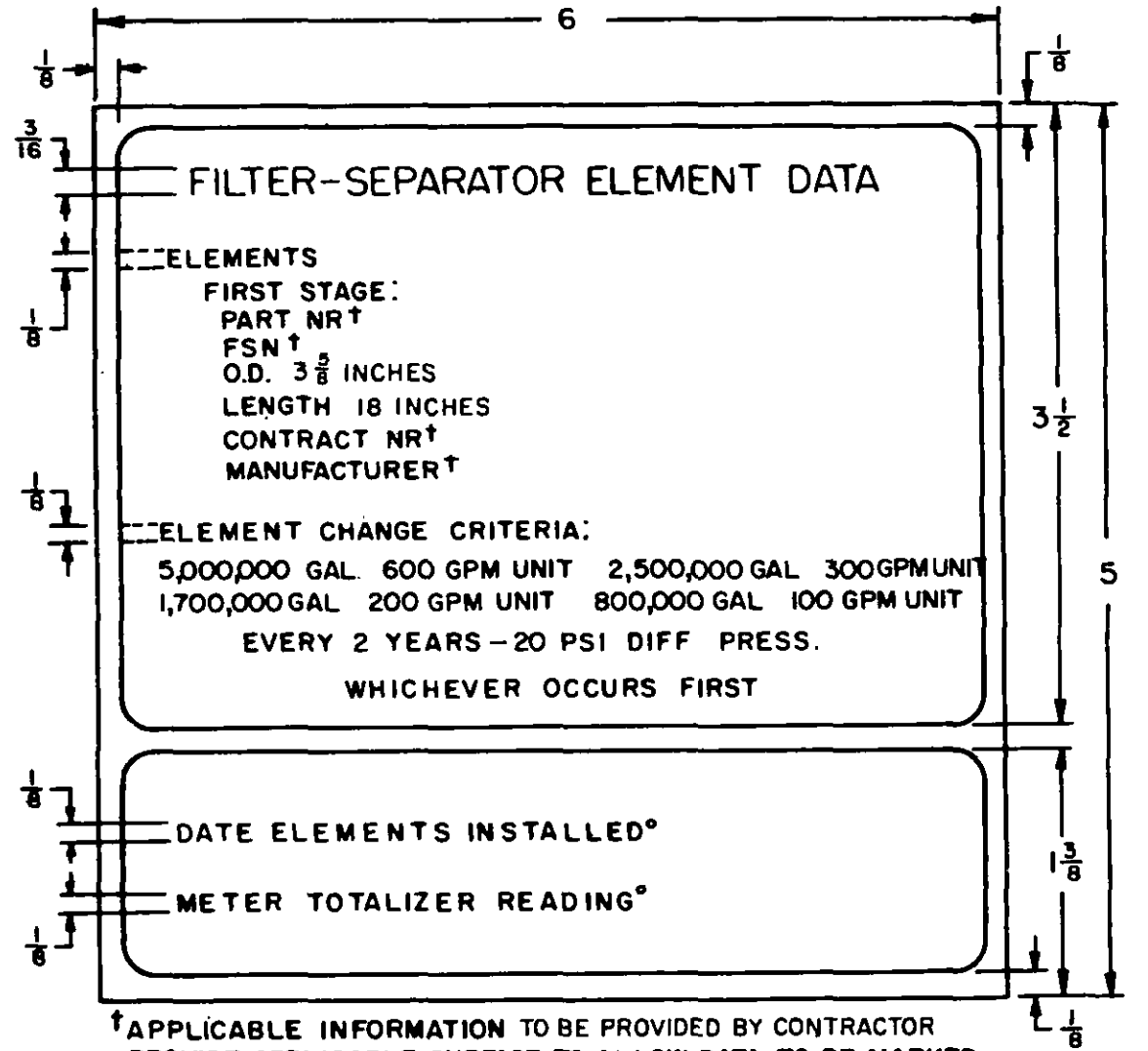
4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection and testing. The inspection and testing of the filter element shall be conducted in a full-scale filter-separator unit in accordance with MIL-F-8901 and as specified herein.

4.2 Test condition. During all applicable full-scale testing, 10 filter elements shall be mounted in a Warner-Lewis Division of Fram Corporation model FCS 105 filter-separator.

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NOTES:

1. ALL LETTERING SHALL BE 3/32-INCH HIGH, UNLESS OTHERWISE NOTED
2. BACKGROUND SHALL BE BLUE
3. DIMENSIONS IN INCHES, TOLERANCE ± 1/32

FIGURE 2. Nameplate

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4.3 Preproduction testing (see 6.2)

4.3.1 Test samples. Unless otherwise specified (see 6.2), the preproduction test samples shall consist of 51 elements representative of a production lot (see 6.3.1). The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. Samples shall be prepared as follows:

4.3.1.1 Data to accompany test samples. The preproduction test samples shall be accompanied by the following data:

a. Brief operating and installation data to enable test personnel to correctly install and operate the equipment

b. Engineering data in the form of assembly drawings (two sets).

4.3.1.2 Test sample and data for procuring activity. When tests are conducted at a site other than the laboratory of the procuring activity, the following shall be furnished that activity:

a. Three copies of a test report in accordance with MIL-STD-831. In addition to results, the report shall contain complete records of the tests, including data sheets, performance curves, chronological test records, photographs, sample calculations, test procedures, and a description of the test apparatus.

b. A sample element removed from the filter-separator which was subjected to the preproduction tests

c. Color photograph of the sample element

d. A new filter element.

4.3.2 Preproduction tests. The preproduction tests shall consist of the applicable tests specified in MIL-F-8901, except as follows:

a. Water removal tests at 10 percent during the third hour are not required

b. During the inhibited fuel test, two elements shall be mounted in parallel and single-pass tests shall be performed at 20 gpm

c. During the full-scale tests, the fuel flow rate of capacity shall be 100 gpm

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d. Sections of the plastic material if used in the construction of the element or specially prepared test specimens shall be tested as follows to verify plastic material compliance. The resultant test data shall be included in the test report.

(1) Tensile strength - ASTM D638, type I

(2) Flexural strength - ASTM D790

(3) IZOD impact strength (notched) - ASTM D256, method A

e. Upon completion of the life test, the filter-separator shall be disassembled and inspected to determine the condition of the filter elements. Defects in the elements such as swelling, damaged elements, and damaged gaskets shall be noted. Excessive swelling of or damage to the elements or other parts shall be cause for rejection.

4.4 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.4.1

b. Sampling plan and test . . . . . See 4.4.2.

4.4.1 Individual tests. Each element shall be examined as required by the examination of product paragraph of MIL-F-8901, and to determine if it conforms to the filter elements design configuration used during preproduction testing.

4.4.2 Sampling plan and test. Samples of the filter elements shall be selected at random from each lot in accordance with MIL-STD-105 and subjected to the filter/coalescer element tests specified in MIL-F-8901. An AQL of 1.5 percent, inspection level I, shall be established. This test shall be considered destructive.

4.5 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.2)

5.1.1 Level A. One filter element shall be packaged in accordance with method III of MIL-P-116 in a container conforming to PPP-B-566, PPP-B-676, or PPP-B-636.

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5.1.1.1 Nameplate. Each nameplate shall be packaged in accordance with method III of MIL-P-116.

5.1.1.2 Intermediate packaging. Ten unit-packaged filter elements and one packaged nameplate shall be intermediate packaged in a container conforming to PPP-B-636.

5.1.2 Level C. Preservation and packaging shall be the minimum necessary to afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing (see 6.2)

5.2.1 Level A. Filter elements shall be packed in overseas-type containers conforming to PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-636, or PPP-B-640. Containers shall be of minimum weight and cube, uniform size and shape, and shall contain like quantities, when practical. Closure of the containers shall be in accordance with the applicable specification and appendix thereto. Containers in accordance with PPP-B-636 shall be sealed at all joints and seams with tape conforming to PPP-T-60.

5.2.2 Level B. The filter element shall be packed as specified in 5.2.1, except containers shall be of the domestic type and closure shall be in accordance with the requirements for domestic containers.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in shipping containers in a manner that will insure safe delivery at the lowest transportation and packing charges. Containers shall meet Consolidated Freight rules and regulations or other rules and regulations applicable to the mode of transportation.

5.3 Shipment marking. In addition to any special marking required by the contract or order (see 6.2) all shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

FILTER ELEMENT, FLUID, PRESSURE MXU-233/E.

6. NOTES

6.1 Intended use. The MXU-233/E filter element is intended for use as a replacement item in ground aviation liquid fuel filtering and servicing systems. This filter element is intended to replace the Warner-Lewis Division of Fram Corporation part No. CC-18 elements.

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6.1.1 Quantity of elements. The quantity of elements used in the Warner-Lewis Division of Fram Corporation filter-separators is as follows:

Model Number	Quantity of Elements
FCS 105	10
FCS 205	20
FCS 245	24
FCS 365	36
FCS 525	52.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for preproduction testing (see 6.2)

c. When the test samples are to be other than as specified in 4.3.1

d. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2)

e. Special shipment marking (see 5.3).

6.3 Definition. For the purpose of this specification, the following definition will apply:

6.3.1 Lot. All filter elements of the same part number manufactured in one production run and submitted for acceptance at the same time shall be considered a lot for acceptance purposes.

6.4 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 84

Preparing activity:
Air Force - 11

Project No. 4330-F107

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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